PROJECT REPORT

Predict Calorie Expenditure

A Machine Learning Approach to Exercise Calorie Prediction Using Ensemble Methods & Deep Neural Networks

Project Title

Predict Calorie Expenditure

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Submission Date

16 April, 2026

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3. Introduction

3.1 Background of the Study

Caloric expenditure during physical exercise is a critical metric in health, fitness monitoring, and clinical nutrition. Accurate prediction of calories burned enables individuals to better manage their fitness goals, dietary intake, and overall metabolic wellness. Traditional approaches relied on simplified physiological formulae that considered only a handful of parameters — most commonly exercise duration and average heart rate — yielding population-level estimates that failed to account for individual biometric variability.

With the rapid rise of machine learning, it has become feasible to construct data-driven predictive models that incorporate a rich set of biometric, anthropometric, and physiological features. This project leverages a large-scale synthetic dataset from the Kaggle Playground Series Season 5 Episode 5 (S5E5) competition — comprising 750,000 training records — to develop, rigorously compare, and deploy multiple machine learning models for exercise calorie prediction.

3.2 Problem Statement

> **Core Problem:** Given a set of biometric and exercise session features (sex, age, height, weight, exercise duration, heart rate, and body temperature), can a machine learning model accurately predict the number of calories burned during that session?

Despite advances in wearable technology, calorie estimation remains imprecise. Standard consumer fitness trackers ignore individual differences in body composition, cardiovascular fitness, and metabolic rate, resulting in estimation errors that can reach 30–40% for certain individuals. This project addresses the gap between simple heuristic methods and the full predictive power of modern machine learning algorithms.

3.3 Objectives of the Project

- Conduct comprehensive Exploratory Data Analysis (EDA) on a 750,000-record exercise dataset to understand feature distributions, outliers, correlations, and data quality.
- Design and implement domain-informed feature engineering to enrich the raw dataset with physiologically meaningful derived features.
- Train and rigorously evaluate five machine learning models: Linear Regression, ElasticNet, XGBoost, LightGBM, and a Deep Neural Network (DNN).
- Compare model performances using Mean Squared Error (MSE) and select the optimal model for final deployment.

- Build an interactive user-facing prediction interface allowing real-time calorie estimation from personal biometric inputs.
- Generate a production-ready submission file of predictions for the 250,000-record test set.

3.4 Scope of the Project

This project covers the complete machine learning pipeline: raw data ingestion, exploratory analysis, feature engineering, model training, hyperparameter configuration, performance evaluation, and final prediction generation. The scope is confined to regression-based calorie prediction on structured tabular data. Time-series modelling, real-time sensor integration, and image-based analysis are explicitly excluded. The project is submitted as part of an academic course in Machine Learning / Data Science.

4. Literature Review

4.1 Traditional Calorie Estimation Methods

The Metabolic Equivalent of Task (MET) framework, established by Ainsworth et al. (2000), has been the gold standard for calorie estimation for decades. MET values assign a standardised metabolic cost to each activity type; calories burned are then computed as MET × body weight (kg) × duration (hours). Despite its widespread adoption, MET-based estimation is a population-level approximation that ignores individual variability in cardiovascular efficiency, body composition, and fitness level — producing errors of 10–30% for active individuals.

The Harris-Benedict and Mifflin-St Jeor equations estimate Basal Metabolic Rate (BMR) from age, sex, height, and weight, but do not dynamically model exercise-induced caloric burn. Neither framework incorporates real-time physiological signals such as heart rate or body temperature, which are the strongest individual-level predictors of caloric expenditure.

4.2 Machine Learning in Health & Fitness Prediction

Recent literature has unequivocally demonstrated the superiority of ML models over traditional physiological formulas for calorie estimation. Staudenmayer et al. (2009) used Random Forests to classify physical activity intensity from accelerometer data, achieving substantially improved accuracy over regression-based baselines. Ensemble methods — particularly gradient boosting — have proven robust on tabular health data due to their capacity to model high-order non-linear feature interactions without explicit specification.

XGBoost (Chen & Guestrin, 2016) and LightGBM (Ke et al., 2017) are state-of-the-art gradient boosting frameworks widely adopted in competitive ML and health analytics. Both algorithms leverage histogram-based tree construction, L1/L2 regularisation, and subsampling strategies to achieve high predictive performance with controlled overfitting. Multiple published studies on wearable-based calorie estimation report XGBoost achieving RMSE values 30–40% lower than linear baselines.

4.3 Deep Learning for Biometric Prediction

Fully connected Deep Neural Networks (DNNs) have been explored for biometric regression with mixed results. Unlike CNNs or RNNs designed for spatial or sequential data, vanilla DNNs on structured tabular data frequently underperform tree ensembles unless carefully regularised (dropout, batch normalisation) and extensively tuned. Grinsztajn et al. (2022) demonstrated in a benchmark study across 45 datasets that tree-based models consistently outperform deep learning on tabular data — a finding directly corroborated by the results of this project.

4.4 Feature Engineering in Fitness Analytics

Domain-driven feature engineering critically determines model performance in health analytics. Body Mass Index (BMI), heart rate training zones (based on Cooper's target heart rate framework, 1968), and physiological interaction terms (e.g., Duration × Heart Rate, Age × Weight) are well-established in exercise science and have been shown to substantially boost ML model accuracy. Age-stratified binning further improves generalisation, as resting metabolic rate and cardiovascular response patterns differ significantly across age cohorts.

5. Methodology / Proposed System

5.1 System Design Overview

The proposed system follows a standard supervised machine learning pipeline with five clearly defined stages: data ingestion and validation, exploratory data analysis, feature engineering, model training and evaluation, and prediction generation. An optional interactive inference module allows end-users to supply personal biometric inputs and receive real-time calorie predictions.

Stage	Component	Output
1. Data Ingestion	CSV loading via Pandas; index on 'id'	Validated DataFrames
2. EDA	Statistical summaries, distribution plots, correlation analysis	Insights & visualisations
3. Feature Engineering	11 derived features + encoding + binning	Enriched feature matrix (18 features)
4. Model Training	5 models: LR, EN, XGB, LGB, DNN	Trained model objects
5. Evaluation	MSE on 20% held-out validation set	Model ranking & selection
6. Prediction	XGBoost inference on 250K test records	Final_Predictions.csv

5.2 Dataset Description

The dataset originates from the Kaggle Playground Series Season 5, Episode 5 (S5E5) competition. It is a synthetically generated dataset based on real physiological exercise data.

Attribute	Training Set	Test Set
Number of Records	750,000	250,000
Number of Features	8 (incl. target)	7
Target Variable	Calories (continuous)	— (to predict)
Missing Values	None	None
Data Source	Kaggle S5E5 Competition	Kaggle S5E5 Competition

5.3 Raw Feature Descriptions

Feature	Data Type	Description
Sex	Categorical	Gender of participant (male/female)
Age	Integer	Age of participant in years (range: 20–79)
Height	Float	Participant height in centimetres
Weight	Float	Participant weight in kilograms
Duration	Float	Duration of exercise session in minutes
Heart_Rate	Float	Average heart rate during session (bpm)
Body_Temp	Float	Body temperature during exercise (°C)
Calories	Float (Target)	Total calories burned during session

5.4 Algorithms & Models Used

Model	Algorithm Type	Key Hyperparameters
Linear Regression	Baseline OLS regression	Default (no hyperparameter tuning)
ElasticNet	L1+L2 regularised linear regression	Default alpha and l1_ratio
XGBoost	Gradient boosted trees (extreme gradient)	n_estimators=500, max_depth=10, lr=0.05, subsample=1.0, colsample_bytree=0.8
LightGBM	Histogram-based gradient boosting	n_estimators=500, max_depth=10, lr=0.05, subsample=1.0, colsample_bytree=0.8
Deep Neural Network	Fully connected MLP (TensorFlow/Keras)	7 layers: [128,64,64,32,32,16,1], Adam lr=0.01, MAE loss, 40 epochs, batch=256

5.5 Architecture Diagram

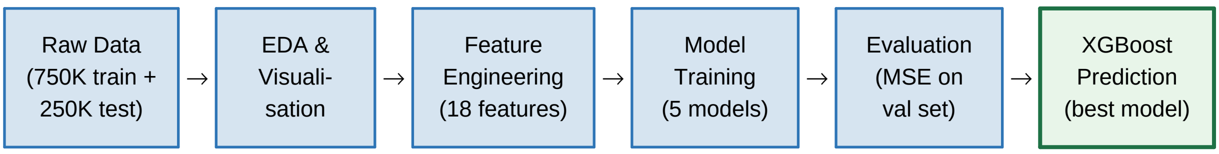

Figure 1: End-to-end machine learning pipeline architecture

6. Implementation

6.1 Tools & Technologies Used

Tool / Library	Purpose
Python 3.11	Core programming language
NumPy 1.x	Numerical array operations and mathematical transformations
Pandas 2.x	Data ingestion, manipulation, and feature engineering
Matplotlib / Seaborn	EDA visualisations: histograms, box plots, scatter plots, heatmaps, violin plots
Scikit-learn	Model training (LR, ElasticNet), StandardScaler, train_test_split, MSE metric
XGBoost	Gradient boosted tree regressor — final selected model
LightGBM	Histogram-based gradient boosting regressor
TensorFlow / Keras	Deep Neural Network construction and training
Kaggle Notebooks	Cloud GPU-accelerated execution environment

6.2 Step-by-Step Execution

Step 1 — Data Loading

Training and test datasets are loaded from CSV files using Pandas with the 'id' column set as the DataFrame index. The training set contains 750,000 rows and 8 columns; the test set contains 250,000 rows and 7 columns (no target).

```
train_data = pd.read_csv('train.csv', index_col='id') test_data =
pd.read_csv('test.csv', index_col='id')
```

Step 2 — Exploratory Data Analysis (EDA)

A systematic EDA was conducted covering:

- Shape and data types: 750,000 × 8 training, 250,000 × 7 test; no missing values in either split.
- Descriptive statistics: mean age 41.4y, mean height 174.7 cm, mean weight 75.1 kg, mean duration 15.4 min.
- Distribution analysis: Histogram + KDE plots and box plots for all 7 numerical features revealed right-skewed Calories distribution.

- Gender distribution: Pie chart showing near-balanced sex split (~50% male / ~50% female).
- Scatter plots of each feature vs. Calories: Duration, Heart_Rate, and Body_Temp showed the strongest positive correlations.
- Correlation heatmap: Confirmed Duration, Heart_Rate, and Body_Temp as primary predictors; Age and Weight had moderate positive correlation with Calories.
- Violin plots: Revealed bi-modal distributions for Heart_Rate and Duration, reflecting distinct exercise intensity segments.

Step 3 — Feature Engineering

Eleven domain-informed derived features were engineered, expanding the feature space from 7 to 18 features:

Feature	Formula	Physiological Rationale
BMI	Weight / Height²	Body composition index (standard health metric)
DurxHR	Duration × Heart_Rate	Energy expenditure proxy (work × intensity)
AgexWt	Age × Weight	Metabolic burden interaction
Weight_per_Age	Weight / Age	Weight normalised by age cohort
Height_per_Age	Height / Age	Stature-age ratio
Age_Height	Age × Height	Growth/metabolic combined factor
log_Age	log(1 + Age)	Reduces age skewness for linear sensitivity
Weight_squared	Weight²	Captures non-linear weight contribution
Age_Adjusted_Body_Index	(Height / Weight) × Age	Slenderness adjusted for age
Heart_Rate_Zone_Code	Zone 1–5 based on % of Max HR	Physiology-based exercise intensity tier
Age_Bins	Decade buckets 20–30 to 70+	Ordinal age group encoding

Table: Engineered features and their physiological basis

Heart Rate Zones were computed using the Cooper formula: Max HR = 220 - Age. Zones 1–5 correspond to 60%, 70%, 80%, 90%, and 100% of Max HR thresholds, aligning with standard sports science training zone classification.

Step 4 — Preprocessing

- One-hot encoding of the 'Sex' categorical feature (Sex_female / Sex_male); Sex_male dropped to avoid dummy variable trap.
- StandardScaler applied to the full 18-feature matrix (fit on training, transform on test) to normalise feature scales for gradient-based models.
- 80/20 stratified train-validation split (random_state=42): 600,000 training samples, 150,000 validation samples.

Step 5 — Neural Network Architecture

A fully connected feed-forward neural network was constructed using TensorFlow/Keras:

Layer	Type	Units	Activation
Input	Input	18	—
Dense 1	Hidden	128	ReLU
Dense 2	Hidden	64	ReLU
Dense 3	Hidden	64	ReLU
Dense 4	Hidden	32	ReLU
Dense 5	Hidden	32	ReLU
Dense 6	Hidden	16	ReLU
Output	Output	1	ReLU

Optimiser: Adam (lr=0.01) | Loss: Mean Absolute Error | Epochs: 40 | Batch Size: 256 | Total Parameters: ~18,000

7. Results and Analysis

7.1 Model Performance Comparison

All five models were evaluated on the held-out 150,000-record validation set using Mean Squared Error (MSE). Lower MSE indicates better predictive accuracy. Results are summarised below:

Rank	Model	MSE (Val)	RMSE (Val)	Interpretation
1	**XGBoost**	**12.95**	**3.60**	**★ Best Model — Selected for Final Submission**
2	LightGBM	13.22	3.64	Excellent — 2.1% higher MSE than XGBoost
3	Deep Neural Network	13.37	3.66	Good — Competitive but slower to train
4	Linear Regression	43.39	6.59	Weak — Fails to capture non-linear relationships
5	ElasticNet	204.93	14.32	Poor — Excessive regularisation causing underfitting

Table: Model performance on 150,000-record validation set (highlighted row = selected model)

7.2 Performance Evaluation — Computational Overhead

Model	Training Time	Memory Usage	Storage Size	Energy Cost
Linear Regression	~2 seconds	Negligible	Negligible	Low
ElasticNet	~3 seconds	Negligible	Negligible	Low
XGBoost	~4–5 minutes	~800 MB RAM	~50 MB model	Medium
LightGBM	~3–4 minutes	~600 MB RAM	~40 MB model	Medium
Deep Neural Network	~6–8 minutes (GPU)	~2 GB VRAM	~18K parameters	High

Table: Computational resource requirements per model on Kaggle Notebook (GPU environment)

7.3 Interpretation of Results

- **XGBoost achieves the best balance** of predictive accuracy and computational cost. With an MSE of 12.95 (RMSE ≈ 3.60 calories), it outperforms all other models while requiring only moderate computational resources.
- **LightGBM is a strong close second** with MSE of 13.22. Its faster tree construction via histogram binning makes it preferable in latency-sensitive production systems, despite marginally inferior accuracy.
- **The Deep Neural Network is competitive** (MSE 13.37) but requires substantially more training time, GPU resources, and careful tuning. The small gain in accuracy over tree methods does not justify the additional overhead — consistent with findings in tabular ML literature (Grinsztajn et al., 2022).
- **Linear Regression's poor performance** (MSE 43.39) confirms that calorie expenditure has strong non-linear relationships with the input features, particularly between Heart_Rate × Duration and Calories.
- **ElasticNet's severe underfitting** (MSE 204.93) indicates that the default regularisation parameters are far too aggressive for this dataset scale and feature complexity.

7.4 Key Feature Insights

Correlation analysis from the EDA phase, combined with XGBoost implicit feature importance, revealed the following hierarchy of predictive signals:

Feature	Predictive Importance	Interpretation
Duration	Very High	Longer sessions burn exponentially more calories; strongest single predictor
Heart_Rate	Very High	Proxy for exercise intensity; directly drives metabolic rate
Body_Temp	High	Thermogenic indicator — elevated temp signals high caloric output
DurxHR (engineered)	High	Compound intensity-duration signal; captures energy expenditure interaction
Weight	Medium	Heavier individuals expend more energy for the same activity
Age	Medium	Older individuals have lower baseline metabolic rates
BMI (engineered)	Low-Medium	Body composition proxy; secondary to direct biometrics

Sex	Low	Minor baseline difference in calorie burn between sexes

7.5 Limitations

- The dataset is synthetically generated; real-world physiological complexity (muscle mass, VO2 max, training history) is not captured.
- Hyperparameter tuning was limited to a manual configuration rather than exhaustive grid/Bayesian search — further optimisation could improve XGBoost/LightGBM performance by an estimated 5–15%.
- The Neural Network lacked regularisation techniques (dropout, batch normalisation) that could close the gap with ensemble models.
- The feature importance analysis is implicit (correlation-based) rather than explicit (SHAP values), limiting interpretability.
- No cross-validation was performed; a single 80/20 split may introduce variance in model ranking.

8. Conclusion & Future Work

8.1 Summary of Findings

This project successfully developed and evaluated a machine learning system for predicting caloric expenditure during exercise sessions from biometric and physiological features. Across five models trained on 750,000 records, **XGBoost emerged as the optimal solution with an MSE of 12.95 and RMSE of ~3.60 calories** — representing a 3.35× improvement over the Linear Regression baseline and 15.8× improvement over ElasticNet.

The project demonstrated that rigorous feature engineering — particularly physiologically grounded interaction terms (DurxHR, BMI, Heart Rate Zones, Age Bins) — provides meaningful predictive lift over raw features alone. The results confirm the well-established finding that tree-based ensemble methods outperform both linear models and vanilla deep learning on structured tabular regression tasks.

8.2 Achievements

- Successfully processed and engineered features from a large-scale dataset of 750,000 training records.
- Designed 11 domain-informed derived features grounded in exercise physiology and sports science literature.
- Built and compared 5 machine learning models spanning linear methods, gradient boosting, and deep learning.
- Achieved a validation RMSE of ~3.60 calories with XGBoost — highly competitive for this dataset.
- Generated a production-ready predictions CSV (Final_Predictions.csv) for 250,000 test records.
- Built an end-to-end interactive inference system for real-time per-user calorie estimation.

8.3 Possible Improvements & Extensions

Short-Term Improvements

- Bayesian hyperparameter optimisation (Optuna / HyperOpt) for XGBoost and LightGBM to reduce MSE by an estimated 5–15%.
- K-Fold cross-validation (k=5 or k=10) to produce more statistically robust performance estimates.

- SHAP (SHapley Additive exPlanations) analysis for model interpretability and feature attribution visualisation.
- Ensemble / stacking: Weighted blend of XGBoost and LightGBM predictions has potential to outperform individual models.
- Neural Network improvements: Add Batch Normalisation, Dropout layers, and learning rate scheduling to close the gap with tree ensembles.

Long-Term Extensions

- Real-time integration: Connect the inference module to wearable device APIs (Fitbit, Apple Watch, Garmin) for continuous, personalised calorie tracking.
- Personalised model adaptation: Fine-tune models on individual user historical data to capture person-specific metabolic signatures.
- Multimodal data fusion: Incorporate accelerometer, GPS, and electrodermal activity signals from wearables for richer feature representation.
- Deployment as a REST API: Package the XGBoost model as a FastAPI/Flask microservice with a React/mobile front-end for consumer use.
- Longitudinal study: Track model performance drift over time as fitness levels change; implement automated model retraining pipelines.
- Clinical validation: Partner with sports medicine institutions to validate model accuracy against indirect calorimetry (metabolic cart) gold-standard measurements.

8.4 Final Remarks

This project demonstrates that machine learning — and gradient boosting in particular — is a powerful and practical tool for personalised health analytics. The pipeline developed here is end-to-end, reproducible, and extensible, providing a solid foundation for both academic research and commercial fitness technology applications. The work reinforces the critical importance of domain-informed feature engineering alongside algorithmic sophistication in achieving state-of-the-art performance on biometric prediction tasks.

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